

Routine Explanation

download_s4g_images.py

This document explains the small downloader routine that reads galaxy names from scrambled_map.txt and downloads corresponding S4G 3.6 micron FITS images from the IRSA public archive.

Big Picture

- Find the default scrambled_map.txt file from the converted bar-profiles project.
- Read galaxy names from the third column of that file.
- Build an IRSA URL for each galaxy using the S4G archive path convention.
- Skip files that already exist locally.
- Download missing FITS files into s4g_images_36um beside the script.

Configuration

DEFAULT_GALAXY_LIST

Points to barprofiles_paper_GB_working_copy/data/scrambled_map.txt, one directory above the downloader package.

DEFAULT_BASE_URL

The root URL for public S4G galaxy data hosted by IRSA/IPAC.

DEFAULT_OUTDIR

The local folder where downloaded FITS files are stored: s4g_images_36um.

SETTINGS

A simple dictionary used instead of command-line arguments. It controls input path, output path, base URL, optional limit, request timeout, and dry-run mode.

Reading Galaxy Names

read_galaxy_names() opens the scrambled map file, skips blank lines and comment lines, splits each remaining line on whitespace, and takes the third field as the galaxy name. The first two fields are ignored here because this downloader only needs the public S4G name.

```
parts = line.split()
if len(parts) >= 3:
    galaxies.append(parts[2])
```

Downloading One Image

`download_image()` handles one galaxy at a time. It constructs the expected S4G FITS URL and matching local output filename. It then follows a simple decision path.

- If the output file already exists, return an Already have message and do not contact the server.
- If `dry_run` is `True`, return the planned URL without downloading anything.
- Otherwise, call `requests.get()` with the configured timeout.
- Treat the response as valid only when the status is 200 and the body is larger than 1000 bytes.
- Write valid content with `outfile.write_bytes()`; otherwise report NOT FOUND with status code and byte count.

```
url = f"{base_url}/{name}/P1/{name}.phot.1.fits"
outfile = outdir / f"{name}.phot.1.fits"
```

Main Routine

`main()` wires everything together. It resolves the configured paths, creates the output directory if needed, reads the galaxy list, optionally truncates it with `SETTINGS["limit"]`, prints a short run summary, then loops over galaxy names and prints the result of each download attempt.

```
galaxies = read_galaxy_names(galaxy_list_file)
if args["limit"] is not None:
    galaxies = galaxies[: args["limit"]]
for name in galaxies:
    print(download_image(...))
```

How To Run Or Test

Run the script directly with Python. For a small test, edit `SETTINGS["limit"]` to a number such as 5. To preview URLs without contacting the server, set `SETTINGS["dry_run"]` to `True`.

```
python download_s4g_images.py
SETTINGS["limit"] = 5      # small test run
SETTINGS["dry_run"] = True # preview only
```

Notable Limitations

- There is no argparse command-line interface; behavior is changed by editing `SETTINGS`.
- A failed request is reported, but the script does not retry transient network failures.
- The >1000 byte check is a lightweight guard against tiny error pages, not a full FITS validation.

- The script assumes the IRSA archive path uses `/ {name} /P1/ {name}.phot.1.fits` for every galaxy.

Summary

In short, this routine is a straightforward cache-aware downloader: it reads target galaxy names, builds predictable IRSA FITS URLs, skips files already present locally, and saves valid-looking FITS responses into the local image directory.